Computer Networks — Transport Layer, Congestion Control & QoS Complete Revision Notes (Hinglish)

UNIT: TRANSPORT LAYER & CONGESTION CONTROL

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1. Transport Layer

OSI Model ki 4th layer hoti hai.

Ye process-to-process delivery provide karti hai.

- Network Layer → Host to Host Delivery
- Transport Layer → Process to Process Delivery

Example:

- Browser → Web Server
- WhatsApp App → WhatsApp Server

Functions of Transport Layer

1. Process-to-Process Delivery

Data ko correct application/process tak pahuchana.

Ye kaam Port Numbers ki help se hota hai.

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2. Segmentation

Large data ko small segments me todta hai.

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3. Multiplexing & Demultiplexing

Multiplexing

Multiple applications ka data combine karna.

Demultiplexing

Receiver side par correct application ko data dena.

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4. Error Control

Errors detect aur recover karna.

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5. Flow Control

Receiver overload na ho isliye sender ki speed control karna.

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6. Connection Control

Connection establish aur terminate karna.

Port Numbers

Application	Port Number
HTTP	80
HTTPS	443
FTP	21
SMTP	25
DNS	53

Transport Layer Protocols

Main protocols:

1. UDP
2. TCP
3. SCTP

2. UDP (User Datagram Protocol)

UDP ek connectionless protocol hai.

Features of UDP

- Connectionless
- Fast
- Unreliable
- No Flow Control
- No Error Recovery

UDP Header

Field	Size
Source Port	16 bit
Destination Port	16 bit
Length	16 bit
Checksum	16 bit

Advantages of UDP

- Fast speed
- Low overhead
- Simple protocol

Disadvantages of UDP

- Packet loss possible
- No delivery guarantee
- No sequencing

Uses of UDP

- Video Streaming

- Online Gaming
- DNS
- VoIP
- Live Broadcast

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3. TCP (Transmission Control Protocol)

TCP ek reliable aur connection-oriented protocol hai.

Features of TCP

- Reliable Delivery
- Error Checking
- Ordered Delivery
- Flow Control
- Congestion Control

TCP Header Fields

Field	Function
Source Port	Sender Application
Destination Port	Receiver Application
Sequence Number	Packet Order
Acknowledgement Number	ACK
Window Size	Flow Control
Checksum	Error Detection
Flags	Control Bits

TCP Flags

Flag	Meaning
SYN	Start Connection
ACK	Acknowledgement
FIN	End Connection

Flag	Meaning
RST	Reset Connection
PSH	Push Data
URG	Urgent Data

TCP Three-Way Handshake

TCP connection establish karne ke liye 3 steps use karta hai.

Step 1: SYN

Client → Server

"Connection establish karna hai"

Step 2: SYN + ACK

Server → Client

"Connection accepted"

Step 3: ACK

Client → Server

"Connection established"

Handshake Diagram

```
Client                         Server

SYN  ------------------------->

      <----------------------  SYN + ACK

ACK  ------------------------->
```

TCP Connection Termination

TCP 4-step termination use karta hai.

```
FIN
ACK
FIN
ACK
```

Flow Control in TCP

TCP Sliding Window mechanism use karta hai.

Receiver batata hai kitna data receive kar sakta hai.

Error Control in TCP

Methods:

- Checksum
- ACK
- Timeout
- Retransmission

UDP vs TCP

Feature	UDP	TCP
Type	Connectionless	Connection-Oriented
Reliability	No	Yes
Speed	Fast	Slow
Error Control	No	Yes
Flow Control	No	Yes
Usage	Gaming, Streaming	Web, Email

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4. SCTP (Stream Control Transmission Protocol)

TCP aur UDP ka advanced version.

Features of SCTP

Reliable

TCP ki tarah reliable.

Message-Oriented

UDP ki tarah message based.

Multi-streaming

Single connection me multiple streams.

Multi-homing

Multiple IP addresses support.

Advantages of SCTP

- Better Reliability
- Less Delay
- Better Fault Tolerance

Uses of SCTP

- Telecom Networks
- VoIP
- Signaling Systems

Connection Management

Connection ko establish, maintain aur terminate karna.

Phases of Connection Management

1. Connection Establishment

TCP → Three-Way Handshake

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2. Data Transfer

Data exchange hota hai.

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3. Connection Termination

TCP → Four-Way Handshake

UNIT: CONGESTION CONTROL

Congestion

Jab network me bahut zyada traffic aa jata hai aur routers packets handle nahi kar paate toh congestion hota hai.

Effects of Congestion

- Packet Loss
- Delay
- Slow Network
- Retransmission

Causes of Congestion

1. High Traffic

2. Slow Routers

3. Limited Bandwidth

4. Retransmission

Congestion Control

Network traffic ko manage karna taaki congestion na ho.

Types:

1. Open Loop Congestion Control
2. Closed Loop Congestion Control

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1. Open Loop Congestion Control

Congestion hone se pehle prevent karne ki technique.

Feedback use nahi hota.

Features

- Prevention Based
- No Feedback
- Simple

Techniques

Retransmission Policy

Unnecessary retransmission avoid.

Window Policy

Proper window size use.

Acknowledgement Policy

ACK packets reduce.

Discarding Policy

Low priority packets discard.

Admission Policy

Extra traffic allow na karna.

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2. Closed Loop Congestion Control

Congestion detect hone ke baad control karna.

Feedback use hota hai.

Steps

1. Congestion Detect
2. Feedback Send
3. Sender Speed Reduce

Methods of Closed Loop Control

Backpressure

Congested node previous node ko slow hone ko bolta hai.

Choke Packet

Router sender ko special control packet bhejta hai.

"Traffic kam karo"

Choke Packet Diagram

```
Sender -----> Router -----> Receiver

            Congestion

Router -----> Choke Packet -----> Sender
```

TCP Congestion Control

TCP congestion control ke liye:

- Congestion Window (cwnd)
- Slow Start
- Congestion Avoidance
- Fast Retransmit
- Fast Recovery

use karta hai.

Congestion Window (cwnd)

Sender kitna data bhej sakta hai without ACK.

Slow Start

Initially small window size se start karta hai.

Growth:

```
1 → 2 → 4 → 8 → 16
```

Exponential increase.

Congestion Avoidance

Threshold ke baad linear increase.

Fast Retransmit

3 duplicate ACK milne par packet immediately retransmit hota hai.

Fast Recovery

Packet loss ke baad speed ko moderate reduce karta hai.

UNIT: QUALITY OF SERVICE (QoS)

Quality of Service (QoS)

Network ki ability to provide:

- Better Speed
- Better Reliability
- Low Delay

QoS Parameters

Parameter	Meaning
Bandwidth	Data Transfer Capacity
Delay	Time Taken
Jitter	Delay Variation
Reliability	Packet Delivery Guarantee

Techniques to Improve QoS

1. Scheduling
2. Traffic Shaping
3. Resource Reservation
4. Admission Control

Important:

- Leaky Bucket Algorithm

- Token Bucket Algorithm

Traffic Shaping

Traffic flow ko smooth banana.

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1. Leaky Bucket Algorithm

Traffic ko constant rate par send karne ki technique.

Working

- Incoming packets bucket me store.
- Fixed rate se packets send.
- Bucket full hone par extra packets discard.

Diagram

```
Incoming Packets
       ↓
   [ Bucket ]
       ↓
 Fixed Rate Output
```

Advantages

- Congestion reduce
- Smooth traffic

Disadvantages

- Burst traffic support nahi karta
- Packet loss ho sakta hai

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2. Token Bucket Algorithm

Leaky Bucket ka improved version.

Working

- Tokens generate hote hain.
- Packet bhejne ke liye token required.
- Tokens available hone par hi packet send hota hai.

Diagram

```
Tokens Generated
       ↓
 [ Token Bucket ]
       ↓
Packets Sent if Token Available
```

Advantages

- Burst traffic allow karta hai
- Better bandwidth utilization

Disadvantages

- More complex

Leaky Bucket vs Token Bucket

Feature	Leaky Bucket	Token Bucket
Output Rate	Fixed	Variable
Burst Traffic	Not Allowed	Allowed
Packet Drop	More	Less
Flexibility	Low	High

Important Terms

Term	Meaning
Segment	TCP Packet
Datagram	UDP Packet
ACK	Acknowledgement
RTT	Round Trip Time
cwnd	Congestion Window

QUICK REVISION

UDP

- Fast
- Unreliable
- Connectionless

TCP

- Reliable
- Connection-Oriented
- Ordered Delivery

SCTP

- Multi-streaming
- Multi-homing
- Reliable

Open Loop

- Prevention Based
- No Feedback

Closed Loop

- Detection + Recovery
- Feedback Based

Leaky Bucket

- Fixed Rate
- Burst not allowed

Token Bucket

- Burst allowed
- Flexible

FINAL MIND MAP

```
Transport Layer
│
├── Functions
│    ├── Segmentation
│    ├── Flow Control
│    ├── Error Control
│    └── Connection Control
│
├── Protocols
│    ├── UDP
│    ├── TCP
│    └── SCTP
│
├── TCP
│    ├── Handshake
│    ├── Flow Control
│    ├── Error Control
│    └── Congestion Control
│
├── Congestion Control
│    ├── Open Loop
│    ├── Closed Loop
│    ├── Choke Packet
│    ├── Slow Start
│    └── Fast Recovery
│
├── QoS
│    ├── Bandwidth
│    ├── Delay
│    ├── Jitter
│    └── Reliability
│
```

```
├── Leaky Bucket
│   └── Fixed Output Rate
│
└── Token Bucket
    └── Burst Traffic Support
```